

DSCI445 - Homework 8

Your Name

Be sure to `set.seed(445)` at the beginning of your homework.

```
#reproducibility
set.seed(445)
```

1. We know that a cubic regression spline with one knot at ξ can be obtained using a basis of the form $x, x^2, x^3, (x - \xi)_+^3$ where $(x - \xi)_+^3 = (x - \xi)^3$ if $x > \xi$ and 0 otherwise. We will now show that a function of the form

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - \xi)_+^3$$

is a cubic regression spline, regardless of the values of $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$.

- a) Find a cubic polynomial

$$f_1(x) = a_1 + b_1 x + c_1 x^2 + d_1 x^3$$

such that $f(x) = f_1(x)$ for all $x \leq \xi$. Express a_1, b_1, c_1, d_1 in terms of $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$.

- b) Find a cubic polynomial

$$f_2(x) = a_2 + b_2 x + c_2 x^2 + d_2 x^3$$

such that $f(x) = f_2(x)$ for all $x > \xi$. Express a_2, b_2, c_2, d_2 in terms of $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$. We have now established that $f(x)$ is a piecewise polynomial.

- c) Show that $f_1(\xi) = f_2(\xi)$. That is, $f(x)$ is continuous at ξ .

- d) Show that $f'_1(\xi) = f'_2(\xi)$. That is, $f'(x)$ is continuous at ξ .

- e) Show that $f''_1(\xi) = f''_2(\xi)$. That is, $f''(x)$ is continuous at ξ .

Hint: Parts (d) and (e) require knowledge of single-variable calculus. As a reminder, given a cubic polynomial

$$f_1(x) = a_1 + b_1 x + c_1 x^2 + d_1 x^3$$

the first derivative takes the form

$$f'_1(x) = b_1 + 2c_1 x + 3d_1 x^2$$

and the second derivative takes the form

$$f''_1(x) = 2c_1 + 6d_1 x$$

2. This question relates to the `College` data set.

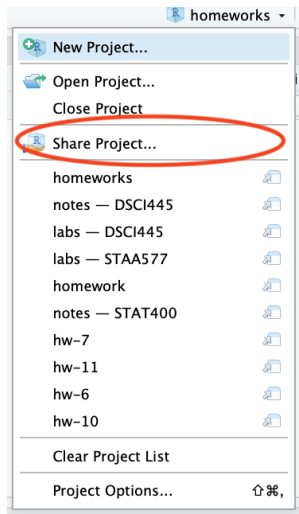
- a) Split the data into a training (60%) and test data set (40%). Using out-of-state tuition as the response and the other variables as the predictors, perform forward stepwise selection on the training set in order to identify a satisfactory model that uses a subset of the predictors.
- b) Fit a GAM on the training data, using out-of-state tuition as the response and the features selected in part a. as predictors with natural cubic splines and $df = 6$ for each continuous variable. Plot the results and explain your findings.
- c) Evaluate the model obtained on the test set and explain the results obtained.

d) For which variables is there evidence of a non-linear relationship with the response?

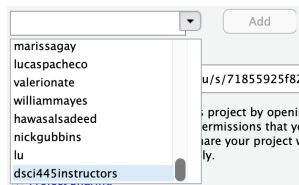
Turn in a pdf of your homework to canvas using the provided Rmd file as a template. Your Rmd file on the server will also be used in grading, so be sure they are identical.

Be sure to share your server project with the instructor and grader. You only need to do this once per semester.

1. Open your **homeworks** project on liberator.stat.colostate.edu
2. Click the drop down on the project (top right side) > Share Project...



3. Click the drop down and add “dsci445instructors” to your project.



This is how you **receive points** for reproducibility on your homework!